

Glossary

IRM: See One Rep Max.

ADENOSINE TRIPHOSPHATE (ATP): A molecule that acts as an energy-transfer medium within each cell. It is one of the end products of cellular metabolism. ATP supplies the energy for muscle contractions when its chemical bonds are broken. ATP is synthesized using the energy released when the macro nutrients in food are metabolized. ATP is continually being broken down and remade via the metabolic processes. The rate at which ATP can be re-synthesized determines the rate of work of which the human body is capable.

AEROBIC BASE: The physiological state in an endurance athlete brought about by extensive training at low-to-moderate intensities. This training enhances the ability of the trained muscles to produce energy aerobically and especially using fat as the primary fuel. This training makes up the vast majority of an endurance athlete's annual training volume. It supports the higher-intensity training by allowing the athlete to recover faster from intense training and races.

AEROBIC DEFICIENCY SYNDROME: A condition common in endurance athletes who spend too much training time doing middle- to high-intensity efforts, which causes an increased development of the anaerobic glycolytic metabolic pathway and reduced development of the basic aerobic metabolic pathway in the affected muscles. If this training state persists long enough, the athlete will see a lowering of his or her Aerobic Threshold.

AEROBIC METABOLISM: In our usage, we are referring to the cellular respiration process that takes place within the mitochondrion and whereby the end products of the breakdown of fat, carbohydrate, and protein are combined with oxygen to produce ATP. It is the primary energy-production pathway for endurance events lasting more than about two minutes.

AEROBIC THRESHOLD (AET): The upper-most intensity of exercise where the production of ATP begins to be dominated

by glycolysis rather than by the oxidation of fats. At this point, blood lactate begins to rise above the resting level. By convention, a blood-lactate concentration of about 2mmol/L (millimole per liter) is indicative of AeT. Another marker of the aerobic threshold is the depth and pace of ventilation. When conversation can't be maintained at a normal cadence, the First Ventilatory Threshold (VT1) has been reached. VT1 corresponds closely to the AeT. This is an important physiological marker of intensity for endurance athletes because it marks the upper level of the most important training zone to use in developing aerobic capacity, the cornerstone to all endurance activities lasting over two minutes. The marker is very trainable and will move higher as measured by both speed of movement and heart rate. Top endurance athletes can have an AeT that is within 10 percent (and less) of their lactate (or anaerobic) threshold.

ANABOLIC: The metabolic process of combining smaller molecules into larger ones. The synthesis of protein that results in new structures within the body is an example of anabolism. Hormones that stimulate the metabolism of protein synthesis are known as anabolic steroids. Training has its effect due to the anabolic process stimulated by the training bouts themselves.

ANAEROBIC METABOLISM: The metabolic chemical reactions that take place without oxygen as one of the reactants. In terms of exercise, it refers to the cellular respiration process that takes place outside the mitochondrion but within the muscle cell, whereby energy is produced to fuel muscle contractions without the use of oxygen. Two types of anaerobic metabolism occur. For very short (ten seconds or less) bouts of very high-intensity exercise, high-energy phosphate fuels stored in the muscle cell as ATP and creatine phosphate (CP) can be used to produce energy for muscle contraction. For longer-duration, high-intensity exercise, the breakdown of glucose through anaerobic glycolysis provides the energy for ATP synthesis. When the requirement for muscle power is greater than can be met with aerobic

glycolysis, the shortfall will be made up by anaerobic glycolysis. Lactate is a byproduct of anaerobic glycolysis and can be used as a marker of intensity of exercise.

ANAEROBIC POWER ENDURANCE: Endurance of maximal effort sustainable for less than sixty seconds. This method comprises a tiny segment of the overall volume of an endurance athlete's annual training. However, it is useful for developing Sport-Specific Strength because it requires the recruitment of a much larger muscle mass than normal, extensive, endurance training.

ANAEROBIC (LACTATE) THRESHOLD (LT): The lowest intensity of exercise at which the production of lactate exceeds the muscle's ability to take up and utilize that lactate as fuel in aerobic metabolism. Above this intensity, lactate levels in the blood begin to rise. The greater the intensity above the LT, the greater the rise in blood lactate.

AUTONOMIC NERVOUS SYSTEM: The part of the nervous system responsible for control of the bodily functions not consciously directed, such as breathing, the heartbeat, and digestive processes.

BASE PERIOD: The training period wherein the athlete focuses on building work capacity in the fundamental components that make up the event being trained for.

BIOGENESIS: The production of new organisms from existing ones. Mitochondria undergo biogenesis as the means of increasing both their size and number within the cells.

BODY WEIGHT: The resistance to movement provided only by a person's own body weight (such as a pull-up or a push-up).

CAPACITY TRAINING: Training that improves the *long-term* performance *potential* of the athlete.

CAPILLARY: The smallest of the body's blood vessels. The cross section of capillaries is on the order of the size of a red blood cell. They transport the blood, along with the nutrients and oxygen that it carries, into intimate contact with the organs and muscles.

CARBOHYDRATE: An organic compound made up only of carbon, hydrogen, and oxygen. In terms of dietary makeup, the carbohydrate group comprises mainly grains, fruits, and starches.

CARDIAC OUTPUT: The amount of blood being pumped by the heart in one minute. It is the product of the stroke volume as measured in liters per beat and the heart rate as measured in beats per minute.

CATABOLIC: The metabolic process of breaking larger molecules into smaller ones for the release of energy. Extensive catabolism of the body's protein structure can have a debilitating effect.

CATALYST: A substance that increases the rate of a chemical reaction by lowering the activation energy required for the reaction to take place. The catalyst is not consumed in the reaction. Biochemical reactions are catalyzed by enzymes.

CIRCUIT: A strength workout where an athlete completes one set each of several exercises in quick succession before returning to the first exercise and repeating the circuit. Can be repeated multiple times with various rest periods as needed for different training effects.

CONTINUITY: Maintaining a regular schedule of training with minimal interruption.

CORE: Vernacular for the musculature of the torso. The core is referred to as the critical link between the shoulder girdle and the pelvis. All athletic movements involve the core in either a static, stabilizing role or in a dynamic role of transmitting motion.

CORE STRENGTH: Strengthening of the core musculature in both its static and dynamic roles is critical for all athletes. Many exercises exist for core strengthening by either isolating the core or by directing forces generated in the limbs through the core to an opposing limb or resistance from gravity.

CREATINE PHOSPHATE: Also known as phosphocreatine, abbreviated PCr or CP, this is a highly bio-available source of high-energy phosphate stored in skeletal muscles. The CP reserve is small, lasting only five to eight seconds of an intense effort. The CP can anaerobically donate a phosphate group to ADP to form ATP for muscular contraction. During periods of low-intensity exercise, excess ATP can be used to resynthesize CP by giving up a phosphate group to the creatine to form CP by a process known as phosphorylation. This continual give-and-take mechanism allows CP to replenish and be available for repeated bouts of high-intensity work.

CYTOSOL: The intracellular fluid in which the various cell structures live. Glycolytic metabolism takes place in the cytosol. Aerobic metabolism takes place within the mitochondria, which in turn lie within the cytosol.

DURATION: The length of time of an exercise bout or training session.

ECONOMY: The energy cost of locomotion. This varies with speed and modality. For example, high running economy on the flats does not directly translate to high running economy while running steeply uphill.

ENDURANCE: The ability to resist fatigue during exercise. For high-intensity exercise, endurance is measured in minutes, whereas in low-intensity exercise, endurance is measured in hours or even days. Duration is inversely proportional to intensity. Proper endurance training helps an athlete increase the time to fatigue at a given intensity.

ENZYME: Biological catalysts responsible for enabling the myriad chemical reactions that sustain life. As with other catalysts, they accelerate the chemical reaction by lowering the activation energy of the reactants and are not consumed during the reaction. In some cases, they speed up the reaction by more than a million times.

FAST-TWITCH (FT) FIBER: A type of muscle fiber that is on the higher power end of the fiber spectrum. These fibers contract more rapidly and with greater force than their slow-twitch relatives. They are larger in cross section, have lower mitochondrial density and lower-density capillary beds suffusing them, and rely more heavily on glycolytic metabolism for ATP production. They have less endurance than the slow-twitch fibers, but a certain class of FT fibers, called FTa, can be trained for more endurance.

FAT (DIETARY): A diverse group of chemical compounds that are insoluble in water. Fats can be categorized as saturated, unsaturated, and trans fats. The chemical bonds in fat allow it to store almost twice the chemical energy per unit of mass than carbohydrates and protein. Because of this, fats provide a large reservoir of energy for low- to moderate-intensity exercise.

FREQUENCY: The number of times each week that an exercise routine is completed.

FUNCTIONAL ADAPTATION: One category of changes to the human body as a result of the recurrent systematic stress of training. The adaptations to training are categorized in two general ways: functional, which relate to the function of the various body systems; and structural, which relate to the body's structures.

GENERALIZED ADAPTATION SYNDROME (GAS): The predictable way the body responds to stress as described by Canadian-Hungarian endocrinologist Hans Selye (1907–1982).

GLYCOLYSIS: The metabolic process that breaks down glucose (a form of sugar derived from carbohydrates) into pyruvate and ATP. Glycolysis is an anaerobic process. The

end product of glycolysis, pyruvate, can enter the mitochondria and undergo further aerobic metabolism if the aerobic capacity is sufficient. Pyruvate can also be a metabolic dead end when the aerobic capacity of the muscle is insufficient. In this case, the metabolite lactate accumulates with negative consequences to endurance. Glycolysis is the primary energy supply for ATP synthesis in high-intensity exercise because glycolysis proceeds at a faster rate than the breakdown of fats.

GRADUALNESS: Adaptation to a training stimulus takes time and occurs in small increments. Rapid increases in training load cannot be accommodated long term. A successful training plan gradually progresses the training load over weeks and months in a systematic way, avoiding large and unmanageable jumps.

HEMOGLOBIN: Abbreviated Hb or Hgb, this is the oxygen-carrying molecule in red blood cells.

HOMEOSTASIS: A stable equilibrium state in an organism. Correct training disrupts homeostasis, resulting in adaptations that allow the body to handle similar training loads better in the future.

HYPERTROPHY: In strength training, this is the training method that induces muscle growth. The increase in muscle volume can be caused by increased sarcoplasmic volume or an increase in the contractile proteins.

INDIVIDUALITY: The need to account for the individual athlete's genetics, training history, lifestyle, etc., when designing training.

INNERVATE: To supply with nerves.

INSULIN: A hormone central to regulating carbohydrate and fat metabolism in the body. Insulin causes cells in the liver, skeletal muscles, and fat tissue to absorb glucose from the blood. In the liver and skeletal muscles, glucose is stored as glycogen, and in fat cells, it is stored as triglycerides.

INTENSITY: Intensity is a measure of the rate of energy consumed by the body. Intensity determines the preferential fuel the muscles use. It also determines the kind of adaptations that will be caused by the training. Common measures of intensity include heart rate, perceived exertion, blood lactate levels, and percentage of VO_2 max.

INTERVAL TRAINING: An endurance training method whereby, within one workout, bouts of higher intensity are separated by rest intervals. This allows the athlete to handle a higher volume of high-intensity training with less fatigue than when doing a continuous high-intensity effort of the same duration.

KREBS CYCLE: Also known as the citric acid cycle, this is one part of the aerobic metabolic process that takes place within a cell's mitochondria and results in the production of ATP.

LACTATE, LACTIC ACID: A chemical product of glycolytic metabolism in muscle cells. It is immediately dissociated into lactate and a hydrogen ion (H⁺). Lactate then has two main pathways available to it whereby it can be used as fuel: 1) It can be converted to pyruvate, which can then enter the Krebs cycle of aerobic metabolism; 2) It can be converted to glucose in the liver by the process of gluconeogenesis. The release of the hydrogen ions can have the effect of lowering the blood's pH (or increasing its acidity). If this continues without adequate buffering, it results in a burning sensation in the muscles and a forced slowing of the pace.

LACTATE BALANCE POINT: The point where lactate production is equal to lactate removal. This is considered the point of maximal intensity that can be maintained for a long duration of many minutes without a subsequent rise in blood lactate level. Also frequently referred to as the lactate threshold (LT), or the maximal lactate steady state (MLSS). This metabolic point has a direct relation to the time to exhaustion at VO₂ max intensity and, as such, bears strongly on the endurance of an athlete. It depends largely on the ability of the active muscles to oxidize lactate. Hence, the aerobic capacity of the slow-twitch muscle fibers, along with the lactate shuttle process, is largely responsible for endurance at high-intensity levels of exercise.

LACTATE SHUTTLE: A mechanism, identified by George Brooks, a professor of integrative biology, in the early 1980s, where lactate is moved out of the muscle cell where it is being produced and into slow-twitch muscle cells where it can enter into the aerobic metabolic pathway to be used as fuel in the synthesis of ATP. This mechanism explains the importance of the Aerobic Base in supporting high-intensity training. The greater the aerobic capacity of the ST fibers to take up and utilize lactate, the more and longer high intensity can be sustained.

LACTATE THRESHOLD (LT) (see Anaerobic Threshold): The highest intensity at which lactate removal is matched by lactate production. Above this intensity lactate begins to accumulate rapidly. Any intensity above the LT will be unsustainable. The higher the intensity the less time it can be sustained.

LIPOLYSIS: The metabolic process of breaking down long-chain lipids or fatty acids into Acetyl CoA that can be used in aerobic metabolism for the synthesis of ATP.

LOAD (WEIGHT): The amount of resistance used. For the transition period, athletes should use a load 50–75 percent of their one-rep maximum. For the max strength period, they should use a load 85–90 percent of their one-rep maximum, or enough to allow only five reps.

LOCAL MUSCULAR ENDURANCE (see Muscular Endurance): The concept of training relatively small muscle groups for endurance without imposing a large load on the cardiovascular system. This effect is accomplished by making the muscular load high through added resistance. This causes the aerobic capabilities of the high-power muscle fibers, which are responsible for the movement, to be the limit on exercise, not the cardiovascular system's ability to supply oxygen to those muscles. The fatigue from this sort of training will be localized to that small group of muscles alone.

MACROCYCLE: From conventional sports, a period representing one complete cycle of training resulting in the preparation for a major event. In competition, this period will usually coincide with an annual cycle of training, hard competition, and the regeneration/recovery necessary before embarking on another annual cycle. It is possible to have two macrocycles in one year.

MAXIMUM LACTATE STEADY STATE: See Lactate Threshold.

MESOCYCLE: A period of training, usually weeks long, during which the training is targeting one or a few physiologic qualities.

METABOLISM: In general usage, this refers to all chemical reactions that occur within an organism. Relative to sport performance, we are mainly interested in the energy production necessary to yield ATP molecules and produce muscular work.

MICROCYCLE: A period of training that is repeated several times during a Mesocycle. Typically one week in length.

MITOCHONDRIA: The tiny (0.5–1 micrometer) organelles within all animal cells responsible for the majority of ATP production. Due to their crucial role in cellular energy production, they are often called the powerhouse of the cell. In exercise, we are mostly interested in the mitochondria in the muscle cells, which undergo adaptation resulting from training.

MODULATION: The variation in training load. Can indicate changes in both day-to-day and week-to-week training loads from light to heavy.

MOTOR UNIT: A group of muscle fibers and the motor nerve that fires them.

MUSCULAR ENDURANCE (ME): Shortened from Local Muscular Endurance (see that definition). The ability to do repeated muscular contractions of the same kind against a resistance. The muscular endurance is not limited by cardiovascular endurance but rather by localized muscular fatigue. In many athletic undertakings, such as running a VK race, the limitations of performance will be largely determined by the local muscular fatigue. As such, the maximal sustainable output, or endurance, is linked directly to the athlete's local muscular endurance in the primary propelling muscle groups.

MUSCLE FIBER: Also known as myocyte or muscle cell, this is an elongated multinucleus cell in human skeletal muscle tissue.

MUSCLE FIBER CONVERSION: Muscle fibers respond directly to the training stimuli imposed upon them. Chronic training of a particular type over many months has the effect of changing the muscle fiber's characteristics. Multi-year studies indicate that prolonged endurance training will result in improved endurance of the faster-twitch fibers within an endurance-trained muscle.

ONE-REP MAX: The maximum load an athlete can lift for one repetition of an exercise movement.

ORGANELLE: A part of the internal cell structure. In our use: Mitochondria.

OVERREACHING: A method used to temporarily apply an excessive training load to the athlete. Can be as short as one to two days or up to a week-long training camp. Needs to be done in a controlled way when the athlete has minimal life stress. Used occasionally by high-level athletes. Must be followed by a significant recovery period.

OVERTRAINING: A condition common in endurance athletes where training bouts are no longer followed by Supercompensation. No matter the length of the recovery window, the athlete's performance degrades. This is a serious condition that needs dramatic intervention.

PARASYMPATHETIC NERVOUS SYSTEM: The part of the involuntary nervous system that serves to slow the heart rate, increase intestinal and glandular activity, and relax the sphincter muscles. Associated with the rest and digest or vegetative states.

PERIODIZE: A structured organization of training used in all modern sports to allow the athlete to focus time and energy on developing one or a few desired training adaptations.

PH: The measure of the acidity or alkalinity of a substance. Used in chemistry and biology to indicate the relative acidity of a compound.

POLARIZED TRAINING: A term used to indicate a training intensity distribution where training volume is dominated by low intensity (80 percent) and high intensity (20 percent) with relatively little moderate intensity; used predominately by elite endurance athletes. This is in contrast to a distribution of intensity that places most volume in the moderate/middle intensity, which is common with lower-level athletes.

POWER: Equals work divided by the time to complete the work. Stated another way: The rate at which work is done. Often confused with energy or work. The crucial differentiating element of power from work or energy is the measure of time in the denominator.

PYRUVATE: One of the end products of glycolytic metabolism that can take one of two metabolic paths: 1) entering the mitochondria and undergoing aerobic metabolism if there is sufficient aerobic capacity in the muscle cell; or 2) accumulating in the cytosol of the cell and being converted to lactate.

RECOVERY TIME: The rest interval between sets of the same exercise or between different exercises when using a circuit or in interval training.

REPETITION OR REP: One complete exercise movement cycle. One pull-up movement would be one rep. One two-minute running bout in an interval workout would be one rep.

SET: A group of repetitions. One set can contain from one rep to many, many reps in the case of muscular endurance training.

SLOW-TWITCH (ST) FIBER: A class of muscle fibers that have greater endurance than their fast-twitch neighbors. The slow-twitch fibers are endowed with more mitochondria, denser capillarization, and higher levels of aerobic enzymes. They are smaller in cross section and contract with less force than the fast-twitch fibers. They have a higher concentration of myoglobin, so appear more red than FT fibers.

SPECIFICITY: Training that mimics the demands of the sport in most ways.

SPORT-SPECIFIC STRENGTH: Training that imposes muscular loads in a sport-specific manner that are higher than those used in the actual event.

STRENGTH TRAINING: Any one of several methods of training directed at improving the contractile qualities of the muscles.

STROKE VOLUME: The volume of blood ejected from the heart with each contraction of the cardiac muscle. This is a highly trainable quality. Endurance training leads to an increase in stroke volume up to a point that is probably largely determined by an athlete's genetics.

STRUCTURAL ADAPTATION: The changes to the body's protein structures brought about by chronic training loads.

SUPERCOMPENSATION: The post-training period during which the parameter that was trained rises to a higher performance capacity than it had before the training.

SYMPATHETIC NERVOUS SYSTEM: A part of the nervous system that serves to accelerate the heart rate, constrict blood vessels, and raise blood pressure. Associated with the fight-or-flight response.

TAPER: To reduce the training load significantly after a training buildup to allow the body to reach a higher level of performance.

TRAINING EFFECT: Indicates, in a qualitative way, how an athlete's body reacts and adapts to the various forms of training stimuli.

TRAINING LOAD: A term used to describe, in a quasi-quantitative way, the type and amount of stress imposed by a single workout or period of training; the Training Load causes the Training Effect. They both depend on the type, the intensity, and the volume of work done in a training bout.

TRAINING STIMULUS: A bout of exercise designed and executed so as to encourage certain adaptations to occur.

TRIGLYCERIDES: A type of fat within the blood that facilitates the transfer of energy either to or from the adipose fat stores of the body. .

UTILIZATION TRAINING: Training that improves the *near-term performance results* of the athlete. Utilization training is commonly prioritized during the buildup to the competition period or the targeted event. This training models the specific demands of the event an athlete is training for.

VENTILATORY THRESHOLD: A notable shift in breathing depth and rate that indicates a change in cellular respiration, and hence metabolism. Useful as a real-time indication of the intensity of exercise. Exercise science recognizes two ventilatory thresholds: VT1, which corresponds to the Aerobic Threshold, and VT2, which corresponds to the Lactate Threshold.

VK: A race involving an elevation gain of one vertical kilometer on steep terrain.

VO₂ MAX: The measure of the maximal aerobic power an athlete can develop. Measured by comparing the rate of inspired and expired oxygen during a multistep exercise test to voluntary exhaustion.

WORK: From physics. Work equals force times the distance over which the force is applied. An example would be that a 50-pound (20kg) weight lifted 10 feet (3m) has had 500-foot pounds (60kg/m) of work done to it.

Index

Page numbers in italic indicate photos.

A

Adamello Ski Raid, 333
 ADP (adenosine diphosphate), 38
 aerobic base development, 48, 156, 158–61, 280–81
 aerobic capacity, 45–46, 76, 90–91, 161, 166–70, 266
 aerobic conditioning, 89–90, 266
 Aerobic Deficiency Syndrome (ADS), 46, 47, 160
 aerobic endurance, 56, 176, 266, 279, 281
 aerobic intervals, 174, 176–78
 aerobic metabolic pathway
 enhancing, 64–65
 metabolic mix and, 39–40
 overview of, 36–37
 protein and, 37
 pyruvate and, 37
 aerobic metabolic system, 26–27
 aerobic power, 92, 266
 Aerobic Threshold (AeT), 42, 44, 46, 63–64, 79, 85, 151, 152–55
 aerobic threshold speed, 89
 aerobic utilization, 77–78
 air squat test, 205–6
 Allen, Mark, 46
 Alps, 14, 29, 31, 340
 Altito-Ternua race, 75
 Ama Dablam, 150, 151
 Ambrose, Caleb, 119
 anaerobic capacity, 77, 266
 anaerobic intervals, 175, 178–83, 185
 anaerobic metabolic pathway (glycolysis)
 enhancing, 64–65
 metabolic mix and, 39–40
 overview of, 36
 pyruvate and, 36, 37
 Anaerobic Threshold. *See* Lactate Threshold
 anaerobic utilization, 78
 Åndalsnes, 22, 23
 antiglycolytic training method, 46, 48
 Arches National Park, 112
 Argentièrre Glacier, 76

Assiniboine, Mount, 161, 162–63
 Åstrand, Per, 170
 ATP (adenosine triphosphate)
 endurance and, 33, 38–39
 function of, 33
 intensity and, 39
 production of, 36–37, 40
 recycling, 36, 38
 attitude, importance of, 82–83, 134–35, 339, 361



Bachelor, Mount, 44
 back squat, 195–96
 bar hang, isometric, 220, 221
 Barkley Marathons, 86, 88, 135
 Base Period
 aerobic base development during, 280–81
 athlete categories for, 285–88
 guidelines for, 302–7
 immediately entering, 269
 individualizing, 278–80, 281
 intensity and, 305–7
 macrocycle plans for, 288–93
 modulation and, 304–5
 overview of, 145
 planning, 281, 285–95, 298–302
 progression and, 304
 purpose of, 277–78
 races during, 278
 strength training during, 281, 307
 weekly vertical in, 305
 week-type menu for, 293–95, 298–302
 base week, 293–94, 322, 323
 Bears Ears National Monument, 164
 Beaufortain, 58–59, 60, 64, 321
 Beccari, Filippo, 125
 Bien, Rod, 51, 142
 Biofava, Federica, 286
 Bollschweiler, Aaron, 217
 Bono, Katie, 78, 80–81

Borysewicz, Eddie, 278
 Boulder, Colorado, 236, 237
 Boulder-White Clouds Range, 165
 Bowman, Bob, 74
 Box, George, 23
 box step-down, 213
 box step test, 208
 box step-up, 212, 245
 brain
 as master control, 25–26, 120
 muscular system and, 27
 Bravo, Verónica, 199
 breaks, returning to training after, 129, 132–33
 Browning, Jeff, 97, 113, 156, 157, 242, 243, 344–45, 346, 355, 356, 357
 Bruchez, Vivian, 76
 Buchs, Emanuel, 283

C

calcium accumulation, 119
 Camonica Valley, 286, 296–97, 298, 346
 Campbell, Jared, 134, 135, 164
 Canova, Renato, 73, 100, 259
 capacity training, 74, 75–76, 78
 carbohydrates
 calories stored in, 24
 as energy source, 55, 60, 62–64
 Castle Valley, 16, 17
 Category 1 vs. Category 2 athletes, 285–88
 Central Governor Theory, 25–26, 120
 Cerro Castillo, 192, 193
 Chase, Anne Gilbert, 197
 chin-ups, 221, 223
 Col du Tourmalet, 75
 commitment, importance of, 261–62
 Competition Period, 146
 Continental Divide Trail, 355
 continuity, 96, 161
 continuous training
 for basic aerobic capacity, 161, 166–70
 higher-intensity, 169–70
 cool-down, 166
 Coots, Forrest, 192, 193
 Coppa Delle Dolomiti, 273
 core strength
 assessment, 226–27, 230
 improving, 225
 usefulness of, 224
 workouts, 231–34
 Cristallo, 228–29, 230
 Critical Pace. *See* Lactate Threshold
 Critical Power. *See* Lactate Threshold
 Critical Speed. *See* Lactate Threshold
 cross-training, 99–100
 Crown Traverse, 186, 187

D

Dallio, Giulia, 298
 Dallio, Riccardo, 298
 Deflorian, Enrico, 262–63
 Delayed Onset of Muscle Soreness (DOMS), 124
 Denali, 14, 78
 de Villa, Javier Martín, 314, 315–16
 diet, 108
 DNF (did not finish), reasons for, 51
 Dolomites, 34–35, 36, 228–29, 230, 241, 254, 255, 273, 285, 295, 336, 337, 361
 Dörschlag, Ekkehard, 186
 Drakensberg Grand Traverse, 206, 207
 durability, importance of, 104
 Duverney, Simon, 130–31, 132

E

Earle, Alex, 256–57, 259
 Eastman, P. D., 357
 electro stim machines, 110, 111
 elevation gain/loss, 127–28
 Elmer, Rico, 282, 283–84
 endurance. *See also* muscular endurance
 ATP's role in, 33
 definition of, 25, 38
 evolution of, 23–25
 fatigue and, 25–27
 metabolism and, 33, 36–45, 62
 strength vs., 193–94
 endurance performance
 factors determining, 27, 32, 33, 45
 improving, 45–46, 48–50, 52–55
 oxygen utilization and, 26–27
 endurance training
 cascade of effects during, 73–74
 counterintuitiveness of, 52–53
 goal of, 25, 26–27, 39, 45
 history of, 71
 philosophy of, 65
 energy
 carbohydrates as source of, 55, 60, 62–64
 fat as source of, 55, 60–64
 low, 124
 Ericsson, Anders, 159
 Etna, Mount, 138, 139
 Everest Marathon, 114–15
 Eysturoy Island, 20–21, 23

F

Farquet, Damien, 282, 283, 284
 fat
 adaptation, 55, 60–61
 calories stored in, 24, 60

as energy source, 55, 60–64

fatigue. *See also* recovery

- brain as arbiter of, 25, 120
- breaks due to, 129
- causes of, 119–20
- embracing, 118
- endurance and, 25–27
- as feedback to training, 102
- frontier fiber model and, 241, 243
- resistance, 27
- short-term, 117–18
- symptoms of, 121–24
- Training Effect and, 73

ferritin, 123

First Ventilatory Threshold (VT1), 79

Flanagan, Tanner, 330

Foote, Mike, 14, 104, 105, 186, 187, 188, 189

Forsberg, Emelie, 12, 274, 360, 361

French, Katie, 194

frontier fiber model, 241, 243

front lunge, 213, 246

Frost, Anna, 114, 115

Frozen Head State Park, 86, 87

Fuji, Mount, 204

Functional Threshold Power or Pace. *See* Lactate Threshold

G

Gallagher, Clare, 200, 201, 254, 255, 295

Gaoligong Mountains, 148–49

gas exchange test, 155

Gates, Rickey, 93, 94–95

Generalized Adaptation Syndrome, 73

Gerschler, Woldemar, 171, 174

Glacier National Park, 68–69, 71, 175, 279

Glen Coe Skyline, 274

glycogen, 24, 36, 48, 60

glycolysis. *See* anaerobic metabolic pathway

Gnam, Alyson Dimmitt, 68–69, 71, 161, 162–63, 175, 194, 279, 303

goal week, 302

Gore-Tex Transalpine Run, 340, 341

gradualness, 96, 98, 117

Grant-Swamp Pass, 134, 242, 243

Green Canyon, 156, 157

Greys River Valley, 308, 309

H

Hagmann, Gregor, 283

Haley, Colin, 76

Handies, Peak, 258, 259

Hardergrat, 130–31, 132, 148

Hardrock 100-Mile Endurance Run, 104, 134, 135, 186, 242, 243, 346, 355

Haute Route, 289, 290

heart, 26

heart rate

- lactate vs., 152–53
- resting, 125

Helgarud, Jan, 75

Hill, A. V., 56, 120

Hill Sprints/Bounding, 179–83, 239–40

Himalaya, 11, 115

hips

- strength assessment for, 204–6, 208–9
- strength training for, 210–14

Hoch Tirol, 290

House, Steve, 13, 15

human growth hormone (HGH), 109

I

illness, returning to training after, 129, 132–33

individuality, 100–101

injuries

- core strength and, 224
- overuse, 144

intensity

- ATP and, 39
- Base Period and, 305–7
- lactate and, 43
- measuring, 79
- metabolic markers of, 79
- zones, 84–85, 89–92

intensity week, 294–95, 298

interval training

- aerobic, 174, 176–78
- anaerobic, 175, 178–83, 185
- high-intensity, 57, 170–71, 174–75, 183, 185
- history of, 171, 174
- principle of, 174
- 30/30s, 177

Irsara, Oskar, 241

J

John Muir Trail, 122

Johnston, Scott, 12, 14, 15, 46, 47, 104, 164, 186, 244, 253, 274

Jones, Dakota, 82, 83

Jornet, Kilian, 10, 11, 12, 13, 14, 15, 22, 23, 66, 67, 70, 71, 72, 76, 79, 89, 90, 102, 104, 106, 109, 110, 118, 126, 127, 136, 138, 158–59, 174, 179, 183, 184, 185, 240, 248, 274, 301, 306–7, 337, 338–39

Juillaguet, Swann, 244

Jungfrau, 224

K

kayaker exercise, 233

Keechelus Ridge, 357

Kelly, John, 86, 87, 88

Khumbu Valley, 150, 151
 Kimmel, Megan, 352
 knee push-ups, 219, 220
 Koerner, Hal, 16, 17
 Krupicka, Anton, 181, 236, 237

L

lactate

- as benefit, 41
- enhanced removal of, 49–50
- heart rate vs., 152–53
- intensity and, 43
- as problem, 41
- production of, 40, 45–46, 48
- shuttle, 49–50

lactate step test, 154–55

Lactate Threshold (LT), 32, 44, 45, 79, 85, 151, 155–56

lactic acid, 40, 45

Lagorai Cima d'Asta Ski Mountaineering Race, 168

Der Lange Weg (The Long Trail), 28, 29, 30, 154

Lanthier, Chloë, 254, 255, 295

Larson, Laura, 186, 189

Latemar Group, 262–63

La Tournette, 83

Leadville Trail 100, 236, 340

legs

- strength assessment for, 204–6, 208–9
- strength training for, 210–14

Le Tour, France, 259

Levy, Jacqui, 303

Lewis Ranch, 175

Little Bighorn Canyon, 357

Lone Peak, 106, 107

Longs Peak, 53

Loowit Trail, 119

Loubet, Remy, 116, 117

Lunger, Tamara, 226

lungs, 26

Lustenberger, Christina, 330

M

Maccabelli, Michele, 34–35, 36, 48

macrocycles, 144–45, 288–93

Maffetone, Phil, 46

Malga Ces Trail, 270

marching plank, 232

Mar Lu Ridge, 86

Marmolada Glacier, 214

massage, 109–11

Matterhorn, 30

Maximum Aerobic Function (MAF), 152–53

maximum anaerobic effort, 96

mechanical economy, 32

mesocycles, 145

metabolic economy, 32

metabolism. *See also* aerobic metabolic pathway; anaerobic metabolic pathway

- definition of, 33, 36

- endurance and, 33, 36–45

microcycles, 145

Miller, Jill, 110

Miró, Mireia, 202, 203

mitochondria, 37, 40

modulation, 96, 98–99

Moehl, Krissy, 147, 148, 149, 258, 259, 267

Montaz-Rosset, Sébastien, 76

Mont Blanc, 76, 109, 190–91, 193, 196, 352. *See also* UTMB

Mont Coin, 190–91, 193

Monte Baldo, 48

Moose Hoofing, 313

Morales, Gerard “Blacky,” 340

Mortirolo Pass, 346

motivation, 137, 200, 339

mountain running

- Category 1 athlete’s training plan for, 342–43, 346–49

- Category 2 athlete’s training plan for, 350–51, 353–54, 358–59

- as complement to skimo and ski mountaineering, 17, 277

- high-intensity workouts for, 306

- training recommendations for, 337–39

- uphill speed for, 277

- weekly vertical for, 358

movement economy, 32, 54–55, 197–98

muscle fibers

- frontier, 241, 243

- slow-twitch vs. fast-twitch, 27, 43, 101

muscles

- activating deep core, 225

- dead, heavy, and flat, 123, 125–26

- sore, 123, 124–25

muscular endurance

- definition of, 178, 240

- determining need for, 240–41, 243

- economy and, 197–98

- gym-based workout, 244–49

- outdoor workout, 249–51

- for skimo, 251–54

- strength and, 198

muscular system, 27

N

Nelson, Luke, 14, 20–21, 23, 106, 107, 156, 164, 165, 188, 254, 255, 268, 269, 274, 275, 280, 295, 308, 309

Néron Ridge, 116, 117

Nice, France, 29, 31

Noakes, Tim, 25, 120

Nolan’s 14, 164–65



'Ohana Trail, 142
 Orgue, Laura, 112
 overtraining, 136–37, 141–44
 overuse injuries, 144
 oxygen
 delivery system, 26
 utilization system, 26–27
 VO2 max, 27, 32, 55, 56–57



Pacific Crest Trail, 355
 Pale di San Martino range, 108
 Palma de Mallorca, 199
 Parigian, Sabina, 53
 Passo del Tonale, 176
 Passo Oclini, 19
 Patience Phase, 46
 Patitucci, Dan, 224
 Patitucci, Janine, 228–29, 230
 Patrouille des Glaciers, 283, 284, 290
 Pazos, Diego, 196
 Pellissier, Jean, 310, 311
 periodization, 144–46
 Pheidippides, 93
 Phelps, Michael, 74
 pick-ups, 168–69
 Pierra Menta race, 58–59, 60, 64, 140, 179, 190–91, 193, 202,
 203, 244, 265, 310, 311, 317, 319, 360, 361
 Pinnacle Peak Trail, 78, 80–81
 Pivk, Tadej, 285
 plank, 226
 marching, 232
 variations, 227
 weighted, 227, 232
 Plaza Geru, 296–97, 298
 Pointe du Dard, 321
 Polarized Training, 92
 Poletti, Michel, 149
 Power of Four, 290
 Precompetition Period, 145–46, 319–20, 324–25
 progressive distance workouts, 169–70
 pull-ups, 196, 221, 223
 assessment, 216
 assisted, 218–19
 lower position for, 216
 Punta Rocca, 34–35, 36
 push-ups
 assessment, 215
 elevated, 221, 222
 inclined, 217–18
 knee, 219, 220
 weighted, 221, 223

Pyrenees, 66, 67, 75, 102, 140, 315
 pyruvate, 36, 37, 40–42, 45



quitting, reasons for, 51



race season
 planning, 263–64, 274–75
 training during, 326–35
 Rainier, Mount, 78, 80–81
 recovery
 enhancing, 106, 108–12
 monitoring for, 117–26
 overtraining and, 137
 times, 102–3, 106
 workouts, 111–12, 166–67
 recovery week, 299–300
 Recovery Zone, 85, 89
 Reiter, Philipp, 30, 154, 171, 226
 resting heart rate, 125
 Rheinwaldhorn, 226
 Rocca, Emma, 340, 341
 Rogers Pass, 330
 Rogotzke, Katie, 68–69, 71
 roller skiing, 251, 252, 253–54
 Roux, Laëtitia, 202, 203, 361
 Rubirola, Marc Pinsach, 138, 139, 140
 Rungger, Raffaella, 241
 Run Rabbit Run 100, 356
 Russell, Mount, 172–73, 174
 The Rut, 274, 280



Saas-Almagell, Switzerland, 239
 Sandes, Ryan, 121, 206, 207
 San Joaquin Couloir, 12
 San Juan Mountains, 97, 135, 259, 275
 Sass Pordoi, 285
 Sawatch Range, 164
 Sayers, Chad, 159
 Sbalbi, Tony, 310, 311
 Schlickenrieder, Peter, 336, 337
 Second Ventilatory Threshold (VT2), 32, 79. *See also* Lactate
 Threshold
 Seiler, Stephen, 92, 170
 self-massage, 110–11
 Selye, Hans, 73
 Shelton, Jenn, 122, 172–73, 174
 Shkhara, Mount, 159
 single leg lateral band, 211
 sit-to-stand test, 209

Ski Bounding, 313

ski ergometer, 253, 254

skimo and ski mountaineering

- assisted pull-ups for, 218–19
- Category 1 athlete's training plan for, 312, 318–20
- Category 2 athlete's training plan for, 321–25
- distinction between, 17, 311
- high-intensity workouts for, 306
- isometric bar hang for, 220, 221
- mountain running as complement to, 17, 277
- pull-up assessment for, 216
- technical aspects of, 317
- training during race season for, 326–35
- uphill speed for, 277
- upper-body training for, 215, 251–54

ski simulations, 313

Ski Striding, 313

sleep, 109

Smiley, Janelle, 28, 29, 31

Smiley, Mark, 29

somatotropin, 109

specificity, 99–100

specificity week, 300–301

speed, 198

split jump squat, 246

squat jump, 247

squats with knee band, 210

stagnation, 96

stair test, 126

standing trunk rotation, 234

step test, 126

St. Helens, Mount, 119, 194

Stofie, Davide, 19

strength. *See also* core strength

- endurance vs., 193–94
- general vs. specific, 194–97
- lower-body, 203
- muscular endurance and, 198
- speed and, 198
- upper-body, 203–4

strength assessment

- core, 226–27, 230
- for legs and hips, 204–6, 208–9
- for upper body, 214–16

strength capacity, 77

strength training

- during Base Period, 281, 307
- core, 225, 231–34
- general, 234–35
- for legs and hips, 210–14
- specific, 239
- for upper body, 217–23

stress, 72, 111, 338

Strom, Kimberly, 150, 151, 238, 239

T

Tahoma, 78

taper week, 301–2

Tarnopolsky, Mark, 110

Täschhorn, 10, 11, 13

Tatoosh Range, 78, 80–81

Tavernaro, Michele, 270

technical capacity, 77

technique, 32, 54–55

tempo workouts, 169

10 percent test, 91, 151, 281

Teocharis, Dimitra, 214

testing, 152–56. *See also* strength assessment

Tetons, 181

30/30s, 177

threshold intervals, 176–77

Tibbetts, Ben, 76

Tortel, Arnaud, 109

training. *See also* Base Period; continuous training; intensity;

- interval training; recovery; Transition Period;
- workouts

- capacity vs. utilization, 74–78
- compulsion and, 132, 137
- continuity in, 96, 97, 161
- cross-, 99–100
- gradualness of, 96, 98, 117
- high- vs. low-intensity, 52–54, 57
- individualistic nature of, 18, 23, 100–101, 259
- mistakes in, 17–18, 101
- modulation of, 96, 98–99
- monitoring, 117–29
- motivation and, 137, 200
- over-, 136–37, 141–44
- periodization of, 144–46
- planning, 259–66
- during race season, 326–35
- returning to, after break, 129, 132–33
- setting goals for, 65, 261, 338–39
- specificity, 99–100
- stress and, 72, 111
- under-, 137

training effect, 73, 126

training load

- definition of, 126
- determining optimal, 158–59
- modulation of, 304–5
- monitoring, 126–29
- progressing, 304
- training effect and, 126

training logs, 126–29

Transcavallo World Cup, 283

Transition Period

- overview of, 145

planning, 272–73
 purpose of, 269
 sample building week in, 271
 sample recovery week in, 271
 starting, 269–70, 272
 weekly aerobic volume for, 271

transverse abdominis (TA), 225, 226

Tre Cime di Lavaredo, 340

Trofeo Mezzalama, 284

Troillet, Florent, 284

Tromsø SkyRace, 274

trunk rotation, standing, 234

U

UltraSignup.com, 274

undertraining, 137

upper body

in skimo, 215

strength assessment for, 214–16

strength training for, 215, 217–23

utilization training, 74–76, 77–78

UTMB (Ultra-Trail du Mont-Blanc), 104, 147–49, 186

V

Val Venegia, 298

Val Veny, 248

Varesco, Christian, 108

Verkhoshansky, Yuri, 46

Vertical Crepa Neigra race, 111, 141, 276, 277, 343

Vienna, Austria, 29

VO2 max, 27, 32, 55, 56–57, 120, 155

W

Wallmann, David, 154, 226

Wapta traverse, 290

warm-ups, 161, 166, 167

Wasatch Mountains, 135

Whitefish Mountain Resort, 186

Whitney, Mount, 122

wildlife, 355, 357

windshield wiper, 230

weighted, 232

Wonderland Trail, 267

workouts. *See also* training

categories of, 266

core strength, 231–34

general strength, 234–35

grading, 128–29

higher-intensity continuous, 169–70

muscular endurance, 244–51

progressive distance, 169–70

recovery, 111–12, 166–67

tempo, 169

Y

Yosemite National Park, 122, 171

Z

Zemmer, Urban, 276, 277, 343

Zone 1 training, 85, 89–90, 266

Zone 2 training, 85, 90–91, 266

Zone 3 training, 85, 91–92, 266

Zone 4 training, 85, 92, 96, 266

Zone 5 training, 85, 96, 266

796.428 JOR

Jornet, Kilian, 1987-
Training for the
uphill athlete : a

09/11/19

Training for the Uphill Athlete Challenging Endurance Sports

SCLSNJ
Hillsborough Branch

SOMERSET COUNTY LIBRARY SYSTEM



3 3665 03317 1464

Like best-selling *Training for the New Alpinism*, *Training for the Uphill Athlete* translates theory into methodology to allow you to write your own training plans and coach yourself to your endurance goals. This is the only book that presents training principles for athletes who regularly participate in distance running, ski mountaineering, skimo, and other sports that require optimum fitness and customized strength.

Trail running, ski mountaineering, skimo, and ultra running are increasingly popular all over the world. This book collects the scientifically backed and athlete-tested wisdom and experience of the best uphill athletes and educates outdoor athletes to develop plans to perform their best.

STEVE HOUSE is a world-renowned climber, mountain guide, and Patagonia ambassador.

SCOTT JOHNSTON coaches several of the nation's top cross-country skiers and was House's coach. Together they own UphillAthlete.com, a coaching site for endurance athletes.

KILIAN JORNET is a professional sky runner, trail runner, ski mountaineer, and ultrarunner. He is a six-time champion of the long-distance Skyrunner World Series and has won some of the most prestigious ultramarathons.

"*Training for the Uphill Athlete* offers ironclad, deliberate, any athlete can hang their hat on. For the serious mountain book is a must."

— NIKKI LAROCHELLE, Member of



296-BAC-529

methodology that
mountaineer, this
mountaineering Team

"This book provides a rigorous, scientific, and individualized ing and coaching for competitive mountain sport endurance athletes and coaches, as well as for beginners."

— Foreword Reviews

patagonia

\$35.00



ISBN: 978-1-938340-84-0



1 92964 41031 2 9 781938 340840